

Measuring a Model of One: A Falsifiable, On-Device Evaluation Methodology for Private Personal Models

Karthikeyan NG*
Independent Researcher

July 2026

Revised August 2026: entity validation on real transcribed diary, three measurement-apparatus faults, and a correction to the zero-egress claim in Section 4.

Abstract

A model built to be private by construction cannot be watched in production: with no egress there is no server-side dashboard, no population A/B test, no confusion matrix, and its one user is the only witness. Such a model must therefore be evaluated where it runs, before it ships, or its fidelity claims cannot be checked at all. This paper’s contribution is a methodology for that setting, and its measurements. TWINEVAL is a falsifiable, baseline-disciplined evaluation harness for on-device personal models: every metric is lift over an explicit baseline; a permutation null and five negative controls collapse to chance when the label-to-text link is destroyed, so the metrics can say no; and validation proceeds in staged legs, synthetic pipeline checks that gate what is worth testing on humans, then population-scale human corpora, then a full-protocol single subject. The method is demonstrated on a deployed testbed, DailyVox, a shipping iPhone voice-journaling app whose verifiable zero-egress architecture is exactly what removes the observability the method has to replace. I lead with what the harness found against its own maker’s model. The deployed model cannot out-forecast persistence at mood prediction; on first contact with a real human diary, my own, its keyword-derived traits scored below a do-nothing baseline (lift -25.2%), a written-versus-spoken register shift its synthetic validation could not expose. At population scale, 1,473 writer-self-labeled diary entries and 15,809 held-out utterances separate the layers that carry real human signal (OS valence, $r = +0.59$) from those that fail (keyword emotion, 43.5% against an 85.0% majority baseline). Learned heads on OS-resident embeddings name the repair, but two heads on two axes give two verdicts: an emotion probe beats the keyword layer on transfer to real diaries (50.1% vs. 43.8%) yet is trained on research-license-only data, and the first license-clean retrain fails the deployment gate by over-firing on neutral text; a separate ~ 105 KB openness head reads 1,307 writers’ self-reported openness at a statistically real but practically worthless $r = +0.13$, bounded by text volume rather than by the linear head. The instrument catching its own model failing is the point: a Personal Model is only trustworthy if its fidelity claims are falsifiable on the device that makes them, and this paper releases the instrument that makes them so.

Keywords: personal model; on-device machine learning; digital twin; digital phenotyping; evaluation methodology; voice journaling; privacy by architecture; negative results; baseline lift.

1 Introduction

The default direction of AI in 2026 is to make the model bigger and the training set wider: one model that reads everyone, improved by the data each person sends it, so the record of who you are becomes the provider’s asset. This paper describes the inverse and argues it is the more important machine for an individual: a *model of one*, trained on a single person, owned by that person, that runs where that person’s data is made and learns only from what they produce.

What makes such a model worth building does not depend on out-predicting anything. Its worth is sovereignty: the only place a person’s most private record can live without a surveillance tax, a model of their own life that they own outright. That worth holds today. The further claim, that a model of one can come to predict and eventually act like the person it is built from, is a bet, not a delivered result, and this paper keeps the two apart from the first page onward.

Privacy by architecture, though, creates the problem this paper is mostly about. A model with no path to transmit its data also cannot be observed at work: no server-side dashboard, no population A/B test,

*Independent researcher. Correspondence: intrepidkarthi@gmail.com. Application source and public research kit: <https://github.com/intrepidkarthi/dailyvox>. App Store: <https://apps.apple.com/app/id6760454642>.

Table 1: The research program at a glance: what is asked, where it stands, what settles it. Verdicts link to the sections that carry them.

Question	Where it stands today	What settles it next
What is a Personal Model, and why build one?	Position stated: five properties; sovereignty over accuracy-supremacy (§2)	—
Is it buildable, privately, today?	Yes — shipped existence proof with verifiable zero egress (§3–4)	—
Can its fidelity claims be falsified?	Yes — metrics collapse under destroyed signal; no permutation of $K=20$ reaches the real result, $p \leq 0.05$ (§6.1)	—
Does the shipped trait layer match a real person’s self-view?	No, measured: -25.2% lift on the author’s diary; register shift root-caused (§6.3)	learned heads (v-next); pilot cohort
Does the analysis layer read real humans’ emotions?	Split: OS valence real ($r=+0.59$); keyword emotion fails (43.5% vs 85.0% baseline; fails by silence) (§6.3)	learned head, license-clean retrain
Do learned heads repair it?	Partly, measured: the probe wins transfer (50.1% vs 43.8%); person-level signal real but thin ($r=+0.13$, $N=1,307$); the first license-clean ship candidate fails the deployment gate — over-fires on neutral text (specificity 32.5% vs 89.5%); within the tested head class, register transfer dominates features (§6.4–6.5)	register-matched, license-clean data (the pilot)
Can the twin forecast mood?	No, measured: loses to persistence (-178%); adopted as a product boundary, not pursued (§6.2)	— (closed)
Does a model of one beat a model of the crowd?	Open — the bet, stated as a bet (§5)	consented pilot cohort (instrument released, §6.6)

no production confusion matrix, and the model’s one user is the only witness. A cloud product earns trust by being watched; a Personal Model, by construction, cannot be. So its fidelity claims are worth something only if they are *falsifiable where the model runs, before it ships*. The central contribution of this paper is a methodology for exactly that setting, and the measurements it produced, most of them negative.

The methodology is TWINEVAL: a falsifiable, baseline-disciplined evaluation harness for on-device personal models, whose reusable core is independent of the app it was built for (Section 6). Every reported number is lift over an explicit baseline, so a model parked at the population mean cannot score; a seeded permutation null and five negative controls collapse to chance when the label-to-text link is destroyed, so the metrics can say no; and validation is staged, from synthetic pipeline checks to population-scale human corpora to a single subject at full protocol. What that discipline surfaced are honest negatives: a deployed model that reads its own maker below a do-nothing baseline (-25.2% lift), a keyword emotion layer that fails on 1,473 real writers while the zero-cost OS valence layer carries real signal ($r = +0.59$), and a learned repair that clears the head-versus-heuristic bar yet fails the deployment gate on the first license-clean retrain.

Three things sit around the methodology to make it necessary and testable, and are supporting rather than central: a *position* (Section 2) that defines the Personal Model and argues its load-bearing property, verifiable zero-egress privacy, is exactly what removes observability; a *substrate and testbed* (Sections 3–4) showing the approach is buildable on shipping 2026 infrastructure and grounding every human number in one deployed artifact, DailyVox; and a *discrimination study* (Section 5) showing a shared model prompted in the author’s voice is separable from his real writing (AUC 0.81), so the path to a Personal Model runs through on-device adaptation [22]. Table 1 states the whole program at a glance: what is asked, where it stands, and what settles it next.

2 The Personal Model

A Personal Model is a model of a single person, owned by that person, that runs where that person’s data is made and learns only from what they produce. Five properties define it, each a constraint rather than a feature. **Of one:** the training set is one life and the target is one person’s next state, not a population average conditioned on a profile. **Private by architecture:** the data never leaves the device, the load-bearing property, a fact about the system’s wiring rather than a policy to trust, and independently verifiable. **On-device:** inference and, over time, adaptation run next to the source, so a small model travels to the data rather than the reverse. **Longitudinal:** the model is assembled from thousands of ordinary days, not one questionnaire afternoon. **Owned:** the weights sit on the person’s side of the trust boundary, an asset rather

than a service rented back.

The contrast with cloud personalization is one sentence: it keeps the model central and pulls the user’s data toward it, while a Personal Model holds the data still and pushes a small model out to it. The variable that moves is egress, and a Personal Model sets it to zero. This is more than on-device inference of a general model: a Personal Model’s subject *is* the user, judged on how well it represents that one person, not on its average over a benchmark population.

3 The substrate, and its drift

The Personal Model is not speculative; as of 2026 the on-device pieces ship in consumer hardware. Apple’s Foundation Models framework exposes a roughly 3-billion-parameter on-device language model with guided generation and small LoRA adapters [5, 1]; on-device transcription has moved to the SpeechAnalyzer stack [3]; a Core AI runtime deploys custom weights on a Neural Engine rated in the tens of TOPS; and on-device personalization from a user’s own data has independent precedent [4]. The substrate for a private model of one is available now.

The platform’s own direction is the more telling signal. At its 2026 developer conference Apple announced that its next-generation foundation models are built with Google’s Gemini and that its assistant is Gemini-powered, routing the heaviest intelligence to a partner’s cloud [2]. The most privacy-committed consumer platform is yielding to the centralizing gravity this paper describes. The Personal Model is defined by resisting it: it stands on the platform’s edge stack while standing against its cloud drift.

4 DailyVox: an existence proof

DailyVox is a deployed iPhone application that builds a Personal Model from voice. A user speaks for roughly forty-two seconds a day; the device transcribes the audio, analyzes it with on-device natural-language tooling, and updates a private model of the user’s communication style, emotional patterns, and a personal knowledge graph [21] of the people and topics in their life. There is no account and no server: during a recording the app issues zero network calls, falsifiable in minutes with a network proxy, and it carries the platform’s strictest “Data Not Collected” privacy designation.¹

The model, which DailyVox calls the Twin, is built in three on-device layers: an extraction layer at write time (per-entry sentiment, entities, topics, and an embedding via `NLTagger` and `NLEmbedding`), a nightly aggregation layer rolling entries into weekly-to-90-day summaries, and an on-demand query layer. What it learns is specific to the person rather than a trait taxonomy: not whether the user is “extroverted,” but that they mention work stress three times more often on Sundays. The strict posture has a cost, no analytics, no crash reporter, no accounts, roughly five times the engineering of a cloud build, but it removes per-request cost entirely and runs in 200–500 ms, so zero-egress is both a privacy stance and the economic precondition for an app that is private and free at scale. DailyVox is the artifact behind the argument, not a thought experiment: a private, on-device, single-person model built from voice with verifiable zero-egress.

5 What is not yet shown

A second limitation is nearer to home than the sampling one. The instrument in this paper measures a model’s *fidelity*; it does not measure whether the deployment around that model keeps the promise the model’s privacy rests on. The transcription defect described in Section 4 passed every fidelity check reported here, because the audio still became the right text and the Twin still folded it correctly — the text simply took a detour through a server on the way. An evaluation harness that cannot see that is incomplete for the setting it was built for, and extending it to assert egress properties as part of the same run is the most concrete piece of future work in this paper.

¹This sentence was false when first written, and the correction belongs in the record. Until version 1.11 (August 2026) the transcriber requested on-device recognition *only when the device was offline*; on a connected phone — nearly every user, nearly always — the audio was sent to the platform’s speech servers for better punctuation. The proxy test described here would have exposed it at any point in the preceding months, and nobody ran it. The defect is now fixed and on-device recognition is unconditional: where no on-device model is installed, transcription fails and says so rather than reaching for the network. I record it because the paper’s argument is that a privately-built model cannot be watched in production, and this is what that costs: a load-bearing privacy claim can be quietly false for months inside an artifact whose author believes it, and only an audit — not a user, not a dashboard, not the store’s own review — can find out.

I am the builder and the subject: I have lived inside my own Personal Model for months, with a twenty-year written diary to hold it against and the source of the model that reads me. That is a deep view of one person, not yet a demonstration that a model of one predicts a person better than a model of the crowd. The position is not a rationalization assembled after the product; I have argued it in public for years.²

To make the question testable, I built an instrument that reproduces DailyVox’s on-device pipeline on a personal computer and ran a discrimination study against an eighteen-year, $\sim 124,000$ -word corpus of my own writing. A stylometric voiceprint (Burrows’s Delta over the 150 most frequent words [18]) is fit on a training split; held-out 200-word passages of my writing define the target. I score three sources against it by area under the ROC curve (1.0 trivially distinguishable, 0.5 indistinguishable): the on-device model prompted to write in my voice, the same model unconditioned, and a word-level model fit directly to my corpus. The result restates the position as a finding. The prompted shared model is reliably distinguishable from me (AUC 0.81, non-overlapping 95% CIs) and being shown my writing barely helps; the model fit to me is not (AUC 0.43). Prompting a shared model with a person’s data is not a model of that person; the fidelity this paper is after comes from fitting to the person. The fitted model here is a word-level stand-in for the on-device LoRA adaptation the paper argues for, and the measure is stylometric, so this is evidence for the mechanism at a single subject, not a finished twin.

I therefore do not rest the case on out-predicting the crowd by a margin of accuracy, an open question the affect-forecasting literature leaves genuinely mixed [30, 31, 32]. I rest it on three things the crowd model cannot offer at once: the model is owned, private by construction, and built in depth on the one person the user cares about. Section 6 reports how far the predictive question has been settled, including where the answer, so far, is no.

6 The instrument, and what it measured

A Personal Model has a validation problem no cloud product has: by construction, no one can watch it work, so its fidelity claims must be falsifiable with instruments that run where the model runs. The measurement gap is not mine alone. The strongest published fidelity results for models of specific people all come from server-side frontier LLMs, interview-based generative agents [6], the Twin-2K-500 program [7], and persona-simulation benchmarks [8]; the academic human-digital-twin field is small, physiology-first, and cloud-based [9]; and the crowded memory-augmented-agent literature [23, 24, 25] evaluates on synthetic personas and simulated dialogues, none of it adapting a model over years of one real person’s record. As far as I can find, no published system combines a longitudinal personal model built from the user’s own journal, fully on-device execution, and a published self-prediction evaluation; this paper supplies that missing evaluation, importing the server systems’ discipline, explicit baselines and test–retest normalization [6], whole.

The harness, TWINEVAL, is a developer-only macOS target in the engine package. It never ships; it replays corpora through the exact production entry point on hermetic in-memory state with fixed seeds and reference dates, so every suite is deterministic. Eight suites run on synthetic or template corpora, FIDELITY, OOC-DRIFT, FORECAST, GROUNDEDNESS, PERSONALITY, MATURATION, MULTILINGUAL, and NEGATIVE-CONTROLS (Section 6.2), alongside the human suites, HUMAN, EMOTION, and the EMOTION-HEAD experiment (Sections 6.3–6.4).

6.1 The discipline

The reusable core of the methodology is two rules, stated once here and applied to every number in the paper. They are what let a private model that no one can watch be evaluated in a way another researcher can trust and rerun.

Reported score = lift over an explicit baseline. For a metric m , a model score s , and a baseline score b_0 (a constant-prior dummy, climatology or persistence for forecasting, a population-mean floor for inference, or the majority class), the reported figure is lift = $(s - b_0)/|b_0|$ (or $s - b_0$ where an absolute scale is meaningful). A model parked at the population mean therefore scores zero, and raw agreement is never reported alone.

Falsifiability null. For any positive result, destroy the label-to-text link (shuffle labels or corpora) across K seeded permutations. The result is trusted only if no permutation reaches it; the empirical p -value is $(c + 1)/(K + 1)$ for c permutations that match or exceed the real score, and every

²Essays at <https://intrepidkarthi.com>, the through-line of roughly 160 written since 2007.

Table 2: Negative-control suite: every metric must collapse when the signal is destroyed. All five controls passed.

Control	Metric	Real	Collapsed
Shuffled-label (fidelity): score each persona against another persona’s label	tracking r (exact-ratio)	+1.00	−0.08
same control	lift vs. 0.5 dummy	+15.5%	−6.8%
Shuffled-corpus (personality): real label, random bag of others’ entries	OCEAN mean r	+0.77	−0.02
Permuted-label (personality): real entries, labels permuted	OCEAN LOO mean r	+0.77	−0.04
Label-permutation null ($K=20$, seeded)	OCEAN mean r vs. null	+0.77	null μ +0.005
Degenerate baselines	dummy / population-mean lift	—	0.0000 / 0.0000

headline metric must collapse to its baseline under the destroyed-signal controls before any of them is believed when it says yes.

Concretely, every trait in the deployed model initializes at 0.5 and moves by running means, so a do-nothing model scores roughly 75% on naive agreement, which is why raw agreement is never reported alone. The harness validates itself: destroying the label-to-text link on purpose collapses every headline metric to its baseline, and no draw from the seeded label-permutation null approaches the real personality result. Table 2 shows the controls in full. Against the $K=20$ null (mean +0.005, s.d. 0.106), the real head’s mean OCEAN r of +0.77 sits far outside the null’s support, so the empirical p is at the floor a $K=20$ null can state, $(0+1)/(20+1) \approx 0.048$ (a $K=999$ null is a camera-ready item). One nuance I surface rather than hide: the shuffled-label fidelity lift lands slightly *negative* (−6.8%), correct because a label-decorrelated predictor is strictly worse than the constant-0.5 population center. This suite validates the *validator*: it proves the harness would catch a broken model, which is what makes the positive results credible, and by itself makes no human-fidelity claim.

6.2 What the synthetic suites measured

Everything in this subsection is measured on synthetic or template-generated corpora. It validates the *pipeline*—the fold machinery, the metrics, the feasibility envelope, and none of it is a human-fidelity claim.

Two synthetic results carry into the human legs. First, against generated personas whose traits are known by construction [10], the deployed engine recovers its targets at +15.5% lift over the do-nothing baseline and stabilizes after roughly twenty entries. Second, a kilobyte-scale ridge head over the OS-resident sentence embeddings (NLEmbedding, zero shipped bytes) recovers assigned Big Five profiles at $r \approx 0.77$ under leave-one-out, and holds at $r \approx 0.74$ trained on one writing style and tested on a deliberately different one, so it reads person signal rather than a stylistic fingerprint, where the engine’s own style heuristics regressed onto OCEAN score −0.13. The five heads export to a 105 KB `.mlmodel`, so size and latency are not barriers. The same synthetic suites also confirm drift separation ($7.4\times$ between shifted and stable cohorts), a maturation knee near $N=20$ entries, and per-language Spanish heads at $r \approx 0.70$ where English keyword lists die. One caveat is load-bearing and repeated wherever these numbers appear: they sit *above* the human text-to-Big-Five state of the art ($r \approx 0.38$ [11]) only because the labels are clean and synthetic; they demonstrate the pipeline’s feasibility and style-invariance, never human fidelity, and motivate the migration to learned heads that the human legs test.

The groundedness suite parses every “Ask Your Twin” answer into claims and asserts them against engine state (600/600 grounded, 1,705 claims); because the chat is template-based this is largely true by construction, so the informative measurement is tonal honesty, and there the suite found a defect. A stress-triggers template ended “This is actually a good sign!” and misfired on 7 of 15 genuinely negative personas, celebrating an empty trigger map as good news. I rewrote the copy, hardened the audit from one literal string to a celebratory-phrase pattern, added a regression test, and re-measured at 15/15; I report the 53.3% before as well as the 100% after, because reporting only the after would commit the sin the harness exists to catch.

One synthetic negative I keep is forecasting. On an 80/20 chronological split the model beats climatology by +5% but loses to a persistence baseline by −178%: it is a conditional-mean estimator, not a trajectory predictor, consistent with a literature in which persistence is hard to beat. This is not a retreat forced by the result; Section 1 pre-committed to it, separating the sovereignty claim that holds today from the predictive bet that does not, so the forecast number sets a product boundary the thesis already allowed for rather than

Table 3: Lemotif entry-level results (1,473 writer-self-labeled diary entries). Raw accuracies are flattering on this positively skewed corpus (happy 730 / jealous 3), which is why each number is shown beside its baseline.

Measurement	Engine	Baseline	Verdict
valence tracking r ($n=1,459$)	+0.59	−0.00 shuffled-label control	positive: real human signal from the zero-cost OS API
sentiment sign accuracy	80.3%	84.4% always-positive	loses to majority: boundary miscalibrated
dominant-emotion accuracy ($n=1,349$)	43.5%	85.0% always-joy	fails outright; neutral fallback on 47% of entries

moving a goalpost. The model is a mirror, not an oracle, and that line is a measurement.

6.3 Human validation: entry-level legs

Synthetic suites can only validate what synthetic corpora can express; the human legs exist because of that limit, and the first proved the point. The fidelity surface under test is what the app calls Twin Resolution: the model’s estimates of seven style traits, scored as $1 - \text{MAE}$ against the user’s own self-report. Six are deterministic keyword-ratio features over the transcript; these are homegrown style measures, not Big Five.

Leg one: the founder’s diary, an honest negative. The first human validation is a single-subject case study at full protocol. I exported my own live diary, twenty-eight transcribed voice entries across three months, folded it through the shipped engine, and scored its seven trait estimates against my self-report. The model failed: resolution 59.5% against an 84.7% do-nothing baseline, a lift of **−25.2%**, describing its own author *worse* than a model that outputs 0.5 for everything. The mechanism is a domain shift the synthetic validation could not expose: the trait heuristics assume written register, expressiveness from exclamation marks transcribed speech never contains, an abstractness lexicon spoken diaries never hit, and collapse to extremes on voice-first text. The synthetic corpora were authored to those very lexicons, so they validated the pipeline while blind to the mismatch. One human was enough to expose it, and to give replacing the keyword traits with learned embedding heads empirical urgency rather than preference (Section 6.4).

A test–retest normalizer the protocol requires comes not from re-answering my own questionnaire but from the person-level suite of Section 6.7, where 1,307 writers’ repeated self-reports give a measured unit-scale self-consistency of ≈ 0.89 ; normalized against it, 59.5% becomes $\approx 67\%$ and the baseline’s 84.7% becomes $\approx 95\%$, the verdict unchanged, since a denominator cannot turn **−25.2%** positive. The negative is evidence not that the Personal Model is unbuildable, the embedding path recovers profiles where the lexicon path fails, but that the evaluation works: a fidelity instrument owned by the subject, run against the subject, caught the model misreading its own primary input modality before any claim shipped on it. Only numbers and root-cause analysis are published; no diary content leaves my machine (Section 8).

Leg two: 1,473 writers, the Lemotif corpus. The second leg scales the *analysis layer*, the exact per-entry path the app runs, to population size using the Lemotif corpus [13]: 1,473 single-day diary answers, each labeled *by its own writer* with one or more of eighteen emotions, so the ground truth is the writer’s own read. The corpus is MIT-licensed and ships with the harness. The paths under test are the engine’s two per-entry readings: **NLTagger** sentiment feeding a valence baseline [20], and a keyword emotion classifier over Plutchik categories plus neutral [19].

Three results (Table 3). The OS valence layer is real: **NLTagger** sentiment tracks writer-labeled valence at $r = +0.59$ over 1,459 entries, the seeded shuffled-label control collapsing to -0.00 , respectably below the text-to-valence state of the art ($r \approx 0.67$ [17]) and the first *positive human* number in this program. The binary sign classifier loses (80.3% against an 84.4% always-positive baseline). And the keyword dominant-emotion classifier fails outright: 43.5% agreement against an 85.0% always-joy baseline, falling back to “neutral” on 47% of entries because the keyword lists never fire (per-class recall runs from joy 44.9% down to disgust 0.0%).

The suite also caught a shipped defect: the engine selected the dominant emotion via a dictionary maximum whose tie-break followed unordered iteration, so the *same text* could receive a *different* emotion between runs in the shipped app. I fixed it with a stable tie-break; every number here is the deterministic post-fix value, a real production bug no synthetic suite had surfaced.

Leg three: the DailyDialog specificity probe. Every Lemotif entry is emotional by construction, so it cannot measure false fires on neutral text. DailyDialog [14], 15,809 held-out utterances with roughly 83%

labeled no-emotion (annotator labels, a perceived-emotion criterion, used for evaluation only under its non-commercial license), fills that gap. The keyword classifier scores specificity 89.5% but sensitivity only 25.0%: read together against a do-nothing 100%/0% classifier, the shipped keyword layer fails *by silence, not by hallucination*. In every register tested it under-reads real human text while the OS-statistical layer carries real signal.

6.4 The learned-head gate

The three human legs converge on one question: do learned heads over OS-resident embeddings beat the keyword layer on *real* human text? Section 6.2 answered it for personality on synthetic labels; the EMOTION-HEAD experiment answers it for emotion with human labels end to end. The setup is minimal: a linear probe over frozen NLEmbedding plus one-vs-rest ridge, trained on 18,000 balanced tweets from the dair-ai emotion corpus [15] (research-use only; nothing trained on it ships). Within-corpus it reaches 48.4% against 16.7% six-class chance, the expected linear-probe band.

Transfer is the number that matters. Head-to-head on 1,340 held-out Lemotif diaries, the learned head wins, **50.1% against the keyword classifier’s 43.8%**, under structural handicaps (12.4% of its predictions are a “love” class Lemotif cannot score, and it saw no diary text in training). The margin is balance: macro recall around 29% against the keyword layer’s 18%, reading the negative emotions the keyword lists structurally miss. The exception is telling, fear collapses to 1.5% on transfer, because tweet-register fear vocabulary is not diary-register anxiety vocabulary: the same register shift that broke the trait layer on my own diary, now inside a learned model.

So the learned-head *experiment* passes on both axes. That is a different bar from the *deployment* gate, which the first license-clean candidate then fails. Trained on 22,549 GoEmotions rows [16], the head works in-corpus and its raw Lemotif number looks spectacular (79.8%), but it sits below the 85.0% always-joy constant with fear and disgust recall at zero, and on DailyDialog it inverts the keyword layer’s failure, specificity 32.5% against 89.5%, hallucinating an emotion on two-thirds of neutral utterances. A ten-configuration sweep of the two repairs needing no new data (rebalancing $\times$ abstention) never encloses the deployed layer’s operating point. Within-register calibration cannot buy cross-register performance; the register-matched, license-clean corpus this leaves as the only path is the one the consented pilot generates. The finding is scoped to the one model class tested, a linear ridge head on a single frozen embedding, and is confounded by label criterion as well as register.

6.5 A fifth leg: real transcribed diary, with gold labels

Every human number above comes from written corpora. Spoken text transcribed by a phone is a different register — disfluent, occasionally mis-transcribed, and carrying names the transcriber has never seen — so the entity layer was re-measured on it directly: 28 entries and 3,687 words from an actual export, hand-annotated with 111 gold spans (43 person, 56 place, 12 organisation).

The result reverses an earlier reading. An unlabelled comparison on the same text had shown the model-free heuristic and the platform recogniser agreeing on only 21.8% of what they found, which looked like the synthetic suite having flattered the heuristic. With gold labels the disagreement resolves the other way: the heuristic reaches **99.1%** recall at 61.6% precision, against the platform’s 94.6% at 32.5%.

The corpus has a property worth stating plainly rather than as a subgroup: *every one of the 43 person names in it is non-Anglo*. There is no common/uncommon split to compute, because the whole diary is the uncommon side. On that population the heuristic finds **100%** of the names and the learned recogniser finds 97.7% — and the heuristic does so precisely *because* it has no learned prior about what a name looks like, so it has none to be wrong about. A model of one person is disproportionately exposed to this: the crowd’s prior is the wrong prior for an individual whose life is not in the training distribution.

6.6 Three times the measurement was the thing that was broken

The instrument’s discipline is baselines and nulls. It is worth reporting that in three cases the fault was not in the model but in the apparatus measuring it, because each was caught by the harness and each would have produced a confident, publishable, wrong number.

A benchmark that cannot discriminate. A widely-cited long-memory benchmark was adopted as the answer-quality bar and then withdrawn on inspection: 6.4% of its answer key is wrong, its LLM judge accepts 63% of intentionally incorrect answers, 56% of its per-category system comparisons are statistically indistinguishable from noise, and a full-context baseline outscores at least one memory system on it — so it

does not measure memory. Two vendors publicly dispute how to run it, one reporting 65.99% for a system whose authors re-ran it at 75.14%. A bar two teams cannot agree how to execute is not a bar worth tuning against.

A metric that rewarded the worse algorithm. Span precision charges one false positive for a detector that glues junk onto a real name (**Told Sarah**) and several for one that reports the junk separately. That accounting is backwards for a knowledge graph: **Told Sarah** is not merely a false positive but a node that fails to unify with **Sarah**, so it silently costs recall on every future question about her — a cost the span metric cannot see. Fixing the detector *lowered* its reported precision by 0.8 points.

A retriever reading the answer off the sort key. A lexical-only arm scored a perfect 100% on an entity-thread corpus. Both the gold entries and the distractors scored identically on the lexical leg, so every case was decided by the identifier tie-break — and the identifiers had been named **thread-...** and **trap-...**, where **thread** sorts first. Fair expectation under a random tie-break was roughly 29%. The corpus now assigns opaque identifiers after shuffling.

None of the three is a failure of the model under test. All three are failures of the surface used to judge it, and a harness whose whole claim is falsifiability has to be as suspicious of its own instruments as of the thing it is pointed at.

6.7 A fourth leg: person-level validation, and what it costs in text

Every human leg so far scores entries; the thesis is about persons. The newest suite makes the person the unit, using Hippocorpus [12]: 1,307 crowd workers, each with at least two real autobiographical stories and their own Big Five openness self-report. Because repeat workers re-answered, I measured the corpus’s own reliability rather than borrowing it: test–retest $r = +0.69$, almost exactly the published $r \approx 0.70$ anchor.

The question is the model-of-one question in miniature: from a person’s own text, can the embedding head estimate *that person’s* self-reported openness better than the population mean? The answer cuts both ways. The signal is statistically real, held-out $r = +0.13$ under seeded ten-fold cross-validation, no permutation null reaching it, and practically worthless, MAE lift $+0.001$ over the population floor; the shipped style heuristics carry no openness signal at all ($r = -0.06$). What bounds this is text volume, not the linear head: the same head class recovers assigned OCEAN at $r = +0.77$ on synthetic corpora and tracks valence at $r = +0.59$ on real diaries, and the human text-to-Big-Five state of the art is itself only $r \approx 0.38$ [11], so non-linearity is not the missing ingredient at ~ 380 words per person. A personal model earns its personality claims with volume, architecture, and register together, or not at all.

6.8 The protocol, and the test that remains

Field-comparable self-prediction numbers require normalization by test–retest consistency [6], but nobody in this program answers twice, public corpora are one-shot, pilot participants are asked once, and I hold myself to the same rule. The protocol is therefore a single tier with a measured denominator: every fidelity number is reported raw against the constant-0.5 dummy, and any normalized figure is divided by the population-scale self-consistency of Section 6.7 (≈ 0.89), the instrument-proxy substitution stated each time. No number here is own-retest-normalized, because no own-retest exists.

The released instrument makes the protocol runnable by others. The public research kit has two parts: *TwinKit*, a runnable on-device replica of the pipeline plus the stylometric layer of Section 5 (the AUC 0.81 vs. 0.43 study is its first output), and a *pilot kit* packaging the questionnaire, consent, and templates for a small consented cohort ($N = 3\text{--}5$), whose diaries are scored locally, with only per-participant aggregate numerics recorded. No public corpus fills that gap: the Pandemic Journaling Project [26] is controlled-access, clinical speech corpora [27] and term-long sensing studies [28] carry no naturalistic voice diary, so the protocol collects its own rather than borrow one that almost fits. The cohort study, predicting next-day and next-week affect against group-level and hybrid baselines following digital-phenotyping practice [29], is future work. What is no longer future work is the measuring stick, and the first real readings from it.

7 Discussion and limitations

One subject at full protocol. The only full-protocol human result is $N=1$ (me), and I am the developer, the opposite of a blinded subject, though the result being negative limits the usual bias concern. Generalization requires the cohort study, which is why the instrument is released rather than merely described.

Synthetic ceilings are not human ceilings. Every positive number in Section 6.2 lives on clean-labeled synthetic corpora; the OCEAN recovery ($r + 0.77$) exceeding the human state of the art ($r \approx 0.38$) is a property of the labels, not the model, which is why I refuse to state it as a human capability.

The written/spoken register gap is the central open threat. It broke the trait layer on my own diary, dents even the learned head (fear recall 1.5%), and none of the population-scale corpora are transcribed speech, so every entry-level human number here carries an unmeasured register term.

Label criteria differ across legs, writer self-labels, annotator labels, and self-report on the model’s own vocabulary; I state the criterion beside every number and never pool across them.

Closed OS primitives: a portability bound, not a privacy hole. The pipeline is built on Apple’s closed primitives, which can drift across releases, so the harness records the OS version and trusts relative comparisons within one. The load-bearing claim is *zero data egress*, and a closed local primitive with no path to transmit data does not weaken it; what suffers is portability off the Apple stack. Substituting open-source primitives would buy off-iOS reproducibility at the cost of no longer measuring the actual shipped model, the object under study, whereas the methodology is primitive-agnostic and ports to any local stack.

What I will not claim. No item-level accuracy, no psychometric-instrument equivalence, no mood prediction (my own forecast suite is the refutation), and no benchmark equivalence with server-side twins; comparisons stay in the ballpark of relative lift over weak baselines. The model remains a mirror of the reflective self as journaled, not an oracle.

8 Ethics and privacy

A diary is among the most sensitive categories of user-generated content, routinely holding mental-health states, relationships, finances, medical symptoms, and grief. Commercial journaling routinely routes exactly this through cloud pipelines [34], and privacy reviews have flagged journaling and mental-health apps for undisclosed sharing and overbroad retention [33]; federated learning and differential privacy reduce but do not eliminate egress. The system studied here adopts the strictest posture, no egress at all, which is what forces this paper’s methodology.

The founder case study uses my own diary, held in a git-ignored local directory, never quoted or paraphrased, with only aggregate numbers published. No app-user data was used, by the architecture of Section 4 I have none. All corpora are published research datasets used within their licenses: Lemotif (MIT) is redistributable; DailyDialog and dair-ai emotion are held locally, never redistributed, and never used to train shipped artifacts. The future pilot is an informed-consent, no-compensation, local-only study publishing aggregate numerics alone.

9 Conclusion

The personal model is coming, and the question is where it will live. The market default puts it in a data center, shared and rented; this paper argues the more valuable form is the opposite, small, private, local, and owned. Its worth today does not depend on prediction. It rests on sovereignty: a model of one’s own life that has no path off the device, a property no cloud model can match. The predictive claim is a bet stated as a bet, and the measurements say we are not there yet, the deployed model cannot out-forecast persistence, its first contact with a real diary exposed a misreading, and its keyword emotion layer lost to a majority baseline on 1,473 real entries. The same measurements say where the repair lies, the OS valence layer carries genuine signal ($r = +0.59$), the learned-head experiment cleared its bar on both axes while the first shippable candidate did not, and the one tonal defect found was fixed and re-measured.

What has changed is that the measuring stick now exists and has been used, on its own maker first. The plainest version of this position is not a promise that the machine already knows you. It is that the only machine worth letting know you is one you own and can keep, whose claims about you are falsifiable on your own device, and a willingness to publish what the falsification finds, year by year, until the readings say otherwise.

Reproducibility

The methodology and the model under study reproduce on different terms, and it is worth separating them. The *methodology* is primitive-agnostic: the baseline-lift discipline, the permutation null, the five negative

controls, and the staged synthetic-then-human legs make no reference to Apple’s stack and transfer to any local pipeline, so a researcher outside the iOS ecosystem can run the same protocol against their own on-device primitives. The *measurements* are tied to a specific shipped pipeline by design, since the object under study is a deployed model, not a portable one, and reproducing the exact numbers requires the same OS-embedded models at the same version (the harness records the OS version per run for this reason). The harness (TWINEVAL) runs as a macOS target against the production engine entry point with fixed seeds; all suites are deterministic. The public research kit, TwinKit (a runnable pipeline replica plus the stylometric measurement layer) and the pilot kit (questionnaire, consent, and response templates), is available in the application repository at <https://github.com/intrepidkarthi/dailyvox> under **research/**. The twin engine itself is a proprietary library; the application is open source. Licensed evaluation corpora (DailyDialog, dair-ai emotion) are not redistributed and must be obtained from their original sources.

Use of AI assistance

An AI coding and writing assistant was used to help draft and edit the prose of this paper and to build parts of the evaluation harness whose results it reports. The design of the study, the measurements, the interpretation of the results, and every claim made here are the author’s own; the author takes full responsibility for the paper’s content, including its numbers and its errors.

References

- [1] Apple Machine Learning Research. Apple Foundation Models Tech Report 2025. 2025. <https://machinelearning.apple.com/research/apple-foundation-models-tech-report-2025>
- [2] Apple Newsroom. Apple Intelligence at WWDC 2026: on-device and Private Cloud Compute foundation models, with a Gemini-powered assistant option. Apple Newsroom, June 2026 (contemporaneous press coverage: TechCrunch, 9 June 2026). <https://www.apple.com/newsroom/>
- [3] Apple. SpeechAnalyzer / SpeechTranscriber (Speech framework), iOS 26. <https://developer.apple.com/documentation/speech>
- [4] On-Device LLM Personalization from User Data. arXiv:2311.12275. <https://arxiv.org/pdf/2311.12275>
- [5] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen. LoRA: Low-Rank Adaptation of Large Language Models. *ICLR*, 2022.
- [6] J. S. Park, C. Q. Zou, A. Shaw, B. M. Hill, C. Cai, M. R. Morris, R. Willer, P. Liang, M. S. Bernstein. Generative Agent Simulations of 1,000 People. arXiv:2411.10109 (v3, 2026). Interview-based GPT-4o agents: 0.83 normalized GSS self-prediction vs. 0.74 demographics-only; Big Five $r = 0.78$, normalized 0.80 by the participants’ 0.95 test–retest replication correlation. <https://arxiv.org/abs/2411.10109>
- [7] O. Toubia et al. Twin-2K-500: Cloud LLM Digital Twins of 2,058 People from Survey Data. *Marketing Science*, 2025. <https://pubsonline.informs.org/doi/full/10.1287/mksc.2025.0262>
- [8] X. Zhu et al. TwinVoice: A Multi-dimensional Benchmark Towards Digital Twins via LLM Persona Simulation. *Findings of ACL*, 2026. arXiv:2510.25536.
- [9] J. Park, E. Park, D. Lee, S. Kang, T. Lee, H. Hong, S.-J. Lee, H. Kim, Y. R. Park, U. Lee. Human Digital Twins for Pervasive Healthcare: A Scoping Review. *Health Informatics Journal*, 2025.
- [10] H. Jiang, X. Zhang, X. Cao, C. Breazeal, D. Roy, J. Kabbara. PersonaLLM: Investigating the Ability of Large Language Models to Express Personality Traits. *Findings of NAACL*, 2024. The assign-traits-then-measure-back protocol adapted here for synthetic ground truth. <https://aclanthology.org/2024.findings-naacl.229>
- [11] G. Park, H. A. Schwartz, J. C. Eichstaedt, M. L. Kern, M. Kosinski, D. J. Stillwell, L. H. Ungar, M. E. P. Seligman. Automatic Personality Assessment Through Social Media Language. *Journal of Personality and Social Psychology*, 108(6), 2015.
- [12] M. Sap, E. Horvitz, Y. Choi, N. A. Smith, J. Pennebaker. Recollection versus Imagination: Exploring Human Memory and Cognition via Neural Language Models. *ACL*, 2020. doi:10.18653/v1/2020.acl-main.178.
- [13] X. A. Li, D. Parikh. Lemotif: An Affective Visual Journal Using Deep Neural Networks. arXiv:1903.07766, 2019. Dataset (MIT): <https://github.com/xaliceli/lemotif>.
- [14] Y. Li, H. Su, X. Shen, W. Li, Z. Cao, S. Niu. DailyDialog: A Manually Labelled Multi-turn Dialogue Dataset. *IJCNLP*, 2017. License CC BY-NC-SA 4.0 (used here for evaluation only).
- [15] E. Saravia, H.-C. T. Liu, Y.-H. Huang, J. Wu, Y.-S. Chen. CARER: Contextualized Affect Representations for Emotion Recognition. *EMNLP*, 2018. (Basis of the dair-ai emotion dataset; research/educational use.)

- [16] D. Demszky, D. Movshovitz-Attias, J. Ko, A. Cowen, G. Nemade, S. Ravi. GoEmotions: A Dataset of Fine-Grained Emotions. *ACL*, 2020.
- [17] S. Buechel, U. Hahn. EmoBank: Studying the Impact of Annotation Perspective and Representation Format on Dimensional Emotion Analysis. *EACL*, 2017.
- [18] J. Burrows. ‘Delta’: A Measure of Stylistic Difference and a Guide to Likely Authorship. *Literary and Linguistic Computing*, 17(3), 2002.
- [19] R. Plutchik. The Nature of Emotions. *American Scientist*, 89(4), 2001.
- [20] J. A. Russell. A Circumplex Model of Affect. *Journal of Personality and Social Psychology*, 39(6), 1980.
- [21] K. Balog, T. Kenter. Personal Knowledge Graphs: A Research Agenda. *ACM SIGIR Int. Conf. on the Theory of Information Retrieval (ICTIR)*, 2019.
- [22] R. M. French. Catastrophic forgetting in connectionist networks. *Trends in Cognitive Sciences*, 3(4):128–135, 1999.
- [23] C. Packer, S. Wooders, K. Lin, V. Fang, S. G. Patil, I. Stoica, J. E. Gonzalez. MemGPT: Towards LLMs as Operating Systems. arXiv:2310.08560, 2023. <https://arxiv.org/abs/2310.08560>
- [24] AI Persona: Towards Life-long Personalization of LLMs. arXiv:2412.13103, 2024. <https://arxiv.org/abs/2412.13103>
- [25] A. Maharana, D.-H. Lee, S. Tulyakov, M. Bansal, F. Barbieri, Y. Fang. Evaluating Very Long-Term Conversational Memory of LLM Agents. *ACL*, 2024. arXiv:2402.17753. <https://arxiv.org/abs/2402.17753>
- [26] S. S. Willen, K. A. Mason, et al. The Pandemic Journaling Project. *PLOS ONE*, 2025 (data: Qualitative Data Repository, doi:10.5064/F6PXS9ZK, controlled access). <https://doi.org/10.1371/journal.pone.0318397>
- [27] B. M. Bot, C. Suver, E. C. Neto, M. Kellen, A. Klein, C. Bare, M. Doerr, A. Pratap, J. Wilbanks, E. R. Dorsey, S. H. Friend, A. D. Trister. The mPower Study, Parkinson Disease Mobile Data Collected Using ResearchKit. *Scientific Data*, 3, 2016. <https://www.nature.com/articles/sdata201611>
- [28] R. Wang, F. Chen, Z. Chen, T. Li, G. Harari, S. Tignor, X. Zhou, D. Ben-Zeev, A. T. Campbell. StudentLife: Assessing Mental Health, Academic Performance and Behavioral Trends of College Students Using Smartphones. *UbiComp*, 2014.
- [29] J. Torous et al. Digital Phenotyping. *npj Digital Medicine*, 2019. <https://www.nature.com/articles/s41746-019-0166-1>
- [30] Next-day affect prediction from wearable data. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10131982/>
- [31] Transformer forecasting of mental state with per-user personalization. arXiv:2601.03603. <https://arxiv.org/pdf/2601.03603>
- [32] Nomothetic vs. idiographic affect models. arXiv:2204.01689. <https://arxiv.org/pdf/2204.01689>
- [33] Mozilla Foundation. *Privacy Not Included: Mental Health Apps. Annual report, 2024.
- [34] Bloom Built / Automattic. Day One Journal: Product and Security Documentation. Accessed 2026.